LEARNER GUIDE

The angle in a semicircle

Circle Theorems · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

This is the first and cleanest consequence of last lesson's keystone. Let the two points A and B be the ends of a **diameter** — a chord straight through the centre O. Then the arc AB is a **semicircle**, and the angle at the centre ∠AOB is a **straight line = 180°**. Since the angle at the circumference is half the angle at the centre, ∠APB = ½ × 180° = **90°**. So **every triangle drawn on a diameter has a right angle** at its third vertex — and it stays 90° wherever P sits on the semicircle. This is often called **Thales' theorem**. It also works in reverse (the **converse**): if an angle at the circumference is 90°, the chord it stands on must be a diameter.

What you are learning to notice

- The **angle in a semicircle is 90°** — an angle drawn on a diameter is a right angle.
- It is the special case of the keystone: a diameter gives 180° at the centre, halved to 90°.
- The other two angles of the triangle then add to **90°** (angles in a triangle sum to 180°).
- **Converse:** a right angle at the circumference must stand on a **diameter**.

Moving through it, one step at a time

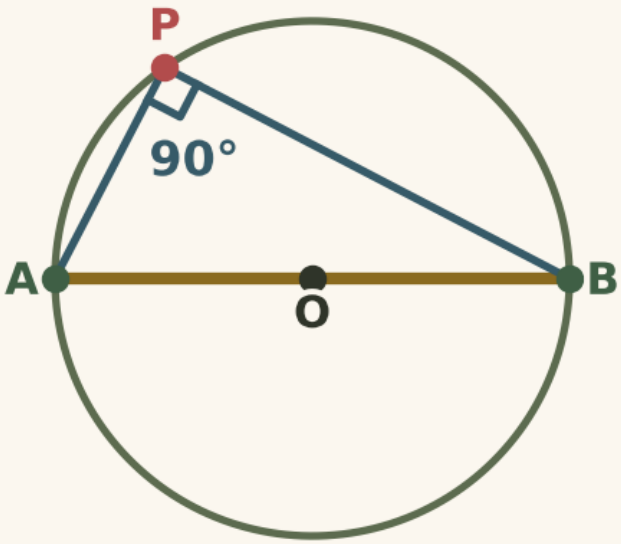
Spot the diameter (a chord through the centre). The angle at the circumference standing on it is 90° — write that with the reason 'angle in a semicircle'. To find another angle in the triangle, use that the three angles sum to 180°, so the two acute angles add to 90°. Always give a reason at each step.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Find the missing angle

AB is a diameter and P is on the circle. One acute angle, ∠PAB, is 35°. Find ∠PBA.

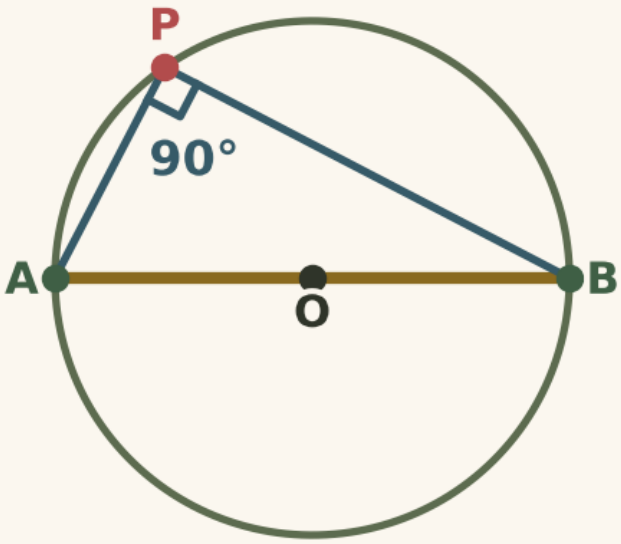


1. AB is a diameter, so ∠APB = 90°.	angle in a semicircle
2. Angles in triangle APB sum to 180°.	35 + 90 + ∠PBA = 180
3. So the two acute angles add to 90°.	∠PBA = 90 – 35

∠PBA = 55°, because the angle in a semicircle is 90° and the angles of the triangle sum to 180°.

Example 2 — Why it is a right angle

Prove that the angle in a semicircle is 90°.



1. AB is a diameter, so it passes through the centre O.	diameter through O
2. A diameter is a straight line, so ∠AOB = 180°.	straight line at O
3. Angle at the centre = 2 × angle at the circumference.	∠APB = ½ × 180°

∠APB = 90°. The angle in a semicircle is a right angle — the special case of the angle-at-the-centre theorem when the arc is a semicircle. ■

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the angle at the centre theorem (Lesson 2), that a diameter passes through the centre, and that angles on a straight line and in a triangle each sum to 180°), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.

- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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